

Polynomials (Part II): Graphs of Polynomial Functions and Finding Polynomials Given Their Zeros

Review on Complex Numbers

Simplify.

- ① $(5+2i)+(-1+3i)$
 $\boxed{4+5i}$
- ② $(-3-5i)+(-2+4i)$
 $\boxed{-5-i}$
- ③ $(-6+3i)-(4-9i)$
 $-6+3i-4+9i$
 $\boxed{-10+12i}$
- ④ $(11+8i)-(-7+i)$
 $11+8i+7-i$
 $\boxed{18+7i}$
- ⑤ $(8-i)+(-6-7i)$
 $\boxed{2-8i}$
- ⑥ $(1+i)+(3-10i)$
 $\boxed{4-9i}$
- ⑦ $(-1-4i)-(2+5i)$
 $-1-4i-2-5i$
 $\boxed{-3-9i}$
- ⑧ $(12-6i)-(-1-6i)$
 $12-6i+1+6i$
 $\boxed{13}$
- ⑨ $3^2(4i) = 12i^2 = 12(-1) = \boxed{-12}$
- ⑩ $-8(-6i) = \boxed{48i}$
- ⑪ $-7i(4i) = -28i^2 = -28(-1) = \boxed{28}$
- ⑫ $5i(7) = \boxed{35i}$

$$\textcircled{13} 3i(-2i) = -6i^2 = -6(-1) = \boxed{6}$$

$$\textcircled{14} (-5i)(-5i) = 25i^2 = 25(-1) = \boxed{-25}$$

- ⑮ $(4-2i)(5+3i)$
FOIL $20+12i-10i-6i^2$
 $20+2i-6(-1)$
 $20+2i+6$
 $\boxed{26+2i}$
- ⑯ $(8-5i)(6-7i)$
 $48-56i-30i+35i^2$
 $48-86i+35(-1)$
 $48-86i-35$
 $\boxed{13-86i}$
- ⑰ $(-2+3i)(-1+10i)$
 $2-20i-3i+30i^2$
 $2-23i+30(-1)$
 $2-23i-30$
 $\boxed{-28-23i}$
- ⑱ $(-1+i)(-3-2i)$
 $3+2i-3i-2i^2$
 $3-i-2(-1)$
 $3-i+2$
 $\boxed{5-i}$
- ⑲ $(7-6i)(7-8i)$
 $49-56i-42i+48i^2$
 $49-98i+48(-1)$
 $49-98i-48$
 $\boxed{1-98i}$
- ⑳ $(-4+i)(-3+2i)$
 $12-8i-3i+2i^2$
 $12-11i+2(-1)$
 $12-11i-2$
 $\boxed{10-11i}$

Finding a Polynomial Given Its Zeros

Leading Coefficient of a Polynomial: The coefficient of the term with the highest power (i.e. greatest degree).

Examples: $2x^4 + 10x^3 - 4x^2 + 9$, $-x^3 + x - 8$, $0.9x^5 + 2x + 4$

$\swarrow$
 $LC = 2$

$\swarrow$
 $LC = -1$

$\swarrow$
 $LC = 0.9$

Write a polynomial function in standard form that has the following zeros, leading coefficient (LC), and degree (D).

- (21) $-1, 2, 4, 1$ $LC = -3$ $D = 4$

$$\begin{aligned} f(x) &= -3(x+1)(x-2)(x-4)(x-1) \\ &= -3(x^2 - 2x + x - 2)(x^2 - x - 4x + 4) \\ &= -3(x^2 - x - 2)(x^2 - 5x + 4) \\ &= -3(x^4 - 5x^3 + 4x^2 - x^3 + 5x^2 - 4x - 2x^2 + 10x - 8) \\ &= -3(x^4 - 6x^3 + 7x^2 + 6x - 8) \\ &= \boxed{-3x^4 + 18x^3 - 21x^2 - 18x + 24} \end{aligned}$$

- (22) $3, -3, -5$ $LC = 1$ $D = 3$

$$\begin{aligned} f(x) &= 1(x-3)(x+3)(x+5) \\ &= \boxed{x^3 + 3x^2 - 9x - 15} \end{aligned}$$

$$= (x^2 - 9)(x + 5)$$

$$= \boxed{x^3 + 5x^2 - 9x - 45}$$

The Complex Conjugate Theorem

If a polynomial has real coefficients and a complex zero $a+bi$, then $a-bi$ must be a zero of the same polynomial.

Write a polynomial function in standard form with real coefficients that has the following zeros, leading coefficient (LC), and degree (D).

- (23) $2i, 6$ $LC = 2$ $D = 3$

$$\begin{aligned} f(x) &= 2(x-2i)(x+2i)(x-6) \\ &= 2(x^2 + 2ix - 2ix - 4i^2)(x-6) \\ &= 2(x^2 - 4(-1))(x-6) \\ &= 2(x^2 + 4)(x-6) \\ &= 2(x^3 - 6x^2 + 4x - 24) \\ &= \boxed{2x^3 - 12x^2 + 8x - 48} \end{aligned}$$

$$\textcircled{24} \quad 3-5i, 1, -2 \quad LC = -2 \quad D = 4$$

$$\begin{aligned} f(x) &= -2(x-(3-5i))(x-(3+5i))(x-1)(x+2) \\ &= -2(x-3+5i)(x-3-5i)(x^2+2x-x-2) \\ &= -2(x^2-3x-5ix-3x+9+15i+5ix-15i-25i^2)(x^2+x-2) \\ &= -2(x^2-6x+9-25(-1))(x^2+x-2) \\ &= -2(x^2-6x+34)(x^2+x-2) \\ &= -2(x^4+x^3-2x^2-6x^3-6x^2+12x+34x^2+34x-68) \\ &= -2(x^4-5x^3+26x^2+66x-68) \\ &= \boxed{-2x^4+10x^3-52x^2-92x+136} \end{aligned}$$

$$\textcircled{25} \quad 1+4i, 4 \quad LC = -1 \quad D = 3$$

$$\begin{aligned} f(x) &= -1(x-(1+4i))(x-(1-4i))(x+4) \\ &= -1(x-1-4i)(x-1+4i)(x+4) \\ &= -1(x^2-x+4ix-x+1-4i^2)(x+4) \\ &= -1(x^2-2x+1-16(-1))(x+4) \\ &= -1(x^2-2x+17)(x+4) \\ &= -1(x^3+4x^2-2x^2-8x+17x+68) \\ &= -1(x^3+2x^2+9x+68) \end{aligned}$$

$$= \boxed{-x^3-2x^2-9x-68}$$

$$\textcircled{26} \quad -i, -1, 2 \quad LC = 3 \quad D = 4$$

$$\begin{aligned} f(x) &= 3(x+i)(x-i)(x+1)(x-2) \\ &= 3(x^2-i^2)(x^2-2x+x-2) \\ &= 3(x^2-(-1))(x^2-x-2) \\ &= 3(x^2+1)(x^2-x-2) \\ &= 3(x^4-x^3-2x^2+x^2-x-2) \\ &= 3(x^4-x^3-x^2-x-2) \\ &= \boxed{3x^4-3x^3-x^2-3x-6} \end{aligned}$$

Graphs of Polynomial Functions

The graphs of polynomial functions move in one direction vertically (either rise or fall) as x -values become very large or very small.

The End Behavior of the Graph of a Polynomial Function: The direction of a graph of a polynomial function as the x -values increase or decrease indefinitely.

The term in a polynomial function with **highest power** determines the end behavior because it grows or decreases faster than all other terms combined.

Example: $2x^5+3x^4-2x^3-6x^2-100x-37$

Notation for Describing End behavior

Notation: $x \rightarrow \infty$

Translation: As x approaches infinity.

Notation: $f(x) \rightarrow \infty$

Translation: $f(x)$ approaches infinity.

Raising a Negative Number to an Even or Odd Power

Even Powers Produce Positive Numbers

I) $(-2)^2 = (-2)(-2) = 4$ II) $(-2)^4 = (-2)(-2)(-2)(-2) = 16$

III) $(-2)^6 = (-2)(-2)(-2)(-2)(-2)(-2) = 64$

Odd Powers Produce Negative Numbers, Assuming that the Base is Negative

IV) $(-2)^3 = (-2)(-2)(-2) = -8$ V) $(-2)^5 = (-2)(-2)(-2)(-2)(-2) = -32$

VI) $(-2)^7 = (-2)(-2)(-2)(-2)(-2)(-2)(-2) = -128$

Indicate the end behavior for each function.

• (27) $f(x) = -2x^6 + 4x^5 - x^4 + x^3 - 20x^2 - 47x + 25$

End Behavior: $\begin{array}{l} \text{As } x \rightarrow \infty, f(x) \rightarrow -\infty \\ \text{As } x \rightarrow -\infty, f(x) \rightarrow -\infty \end{array}$

(28) $f(x) = x^3 - 10x^2 - 2$

End Behavior: $\begin{array}{l} \text{As } x \rightarrow \infty, f(x) \rightarrow \infty \\ \text{As } x \rightarrow -\infty, f(x) \rightarrow -\infty \end{array}$

(29) $f(x) = -3x^5 + 2x^2 - 4x + 6$

End Behavior: $\begin{array}{l} \text{As } x \rightarrow \infty, f(x) \rightarrow -\infty \\ \text{As } x \rightarrow -\infty, f(x) \rightarrow \infty \end{array}$

(30) $f(x) = 9x^4 - 8x^3 - 4x^2 - 71x - 1$

End Behavior: $\begin{array}{l} \text{As } x \rightarrow \infty, f(x) \rightarrow \infty \\ \text{As } x \rightarrow -\infty, f(x) \rightarrow \infty \end{array}$

• (31) $f(x) = -7x^7 + 2x^6 - 3x^3 + 2x$

End Behavior: $\begin{array}{l} \text{As } x \rightarrow \infty, f(x) \rightarrow -\infty \\ \text{As } x \rightarrow -\infty, f(x) \rightarrow \infty \end{array}$

(32) $f(x) = 25x^8 + 9x^7 - 22x^6 + 15x^5 - 4x^3 - 2x + 1$

End Behavior: $\begin{array}{l} \text{As } x \rightarrow \infty, f(x) \rightarrow \infty \\ \text{As } x \rightarrow -\infty, f(x) \rightarrow \infty \end{array}$

33) $f(x) = -4x^4 + 27x - 4$

End Behavior: $\boxed{\begin{array}{l} \text{As } x \rightarrow \infty, f(x) \rightarrow -\infty \\ \text{As } x \rightarrow -\infty, f(x) \rightarrow -\infty \end{array}}$

34) $f(x) = 17x^3 + 2x^2 + x - 10$

End Behavior: $\boxed{\begin{array}{l} \text{As } x \rightarrow \infty, f(x) \rightarrow \infty \\ \text{As } x \rightarrow -\infty, f(x) \rightarrow -\infty \end{array}}$

Describing Where a Function Is Increasing or Decreasing

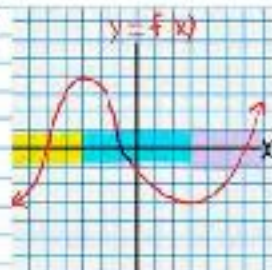
A function is increasing if the y-values increase as the x-values increase.

A function is decreasing if the y-values decrease as the x-values increase.

The graphs of various functions are written below. Write the intervals where each function is increasing or decreasing.

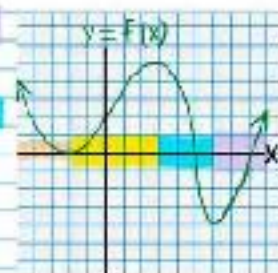
- 35) Increasing: $(-7, -3)$ and $(3, 7)$

Decreasing: $(-3, 3)$



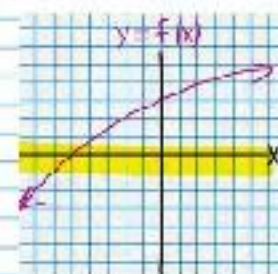
- 36) Increasing: $(-2, 3)$ and $(6, 9)$

Decreasing: $(-5, -2)$ and $(3, 6)$



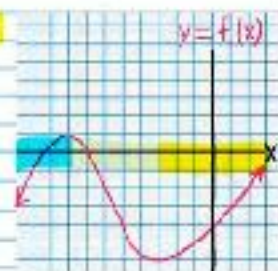
- 37) Increasing: $(-8, 6)$

Decreasing: $\times$



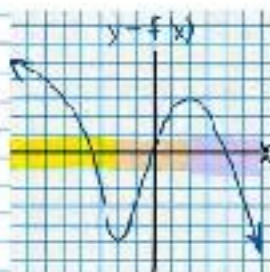
- 38) Increasing: $(-1, -8)$ and $(-3, 3)$

Decreasing: $(-8, -3)$



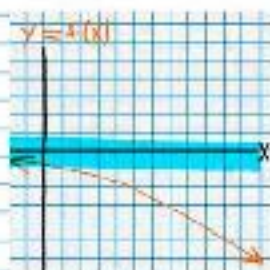
③ Increasing: $(-2, 2)$

Decreasing: $(-8, -2)$ and $(2, 6)$



④ Increasing: x

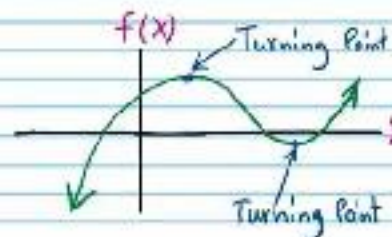
Decreasing: $(-2, 2)$



Turning Point: A point where a function changes from increasing to decreasing or decreasing to increasing.

Example:

$$f(x) = x^3 - 8x^2 + 11x + 20$$



The Turning Points Theorem: The graph of a polynomial function with degree n has at most $n-1$ turning points.

For each polynomial function, state the maximum number of turning points of the corresponding graph.

• ① $f(x) = -3x^5 + 2x^2 - 4x + 6$

Max # of Turning Points: $n-1 = 5-1 = 4$

② $f(x) = 9x^4 - 8x^3 - 4x^2 - 71x - 1$

Max # of Turning Points: $n-1 = 4-1 = 3$

③ $f(x) = -7x^7 + 2x^2 - 3x^3 + 2x$

Max # of Turning Points: $n-1 = 7-1 = 6$

• ④ $f(x) = 25x^8 + 9x^7 - 22x^6 + 15x^5 - 4x^3 - 2x + 1$

Max # of Turning Points: $n-1 = 8-1 = 7$

⑤ $f(x) = -4x^4 + 27x - 4$

Max # of Turning Points: $n-1 = 4-1 = 3$

⑥ $f(x) = 17x^3 + 2x^2 + x - 10$

Max # of Turning Points: $n-1 = 3-1 = 2$

The Multiplicity of a Zero: If a zero is repeated m times, then the multiplicity of the zero is defined to be m .

Example: $f(x) = (x-7)^2(x+2)^3(x-4)$

7 is a zero of multiplicity 2

-2 is a zero of multiplicity 3

4 is a zero of multiplicity 1

Using Multiplicity When Graphing

If a polynomial function has a real zero that has an odd multiplicity, then the graph of the polynomial crosses the x -axis at that number. If a polynomial function has a real zero that has an even multiplicity, then the graph of the polynomial touches the x -axis at that number, but does not pass through it.

Polynomial functions are written below in factored form. Write the x -intercepts of each corresponding graph and determine if the function crosses or touches the x -axis at each intercept.

• (27) $f(x) = 2(x-6)^2(x+4)^3(x-2)$

$6 \Rightarrow$ crosses $2 \Rightarrow$ crosses
 $-4 \Rightarrow$ Touches

By Definition, the Converse of the Factor Theorem is Part of the Factor Theorem: Let f be a polynomial function. If $x-c$ is a factor of $f(x)$, then $f(c)=0$.

(43) $f(x) = -9(x+2)^5(x+1)^4$

$-2 \Rightarrow$ crosses

$-1 \Rightarrow$ Touches

(49) $f(x) = 7(x-8)(x+\sqrt{3})(x+11)^6$

$8 \Rightarrow$ crosses

$-\sqrt{3} \Rightarrow$ crosses

$-11 \Rightarrow$ Touches

• (50) $f(x) = -3(x+5)^7(x-10)^{12}$

$-5 \Rightarrow$ crosses

$10 \Rightarrow$ Touches

51) $f(x) = 53(x-9)^{10}(x+2)(x+5)^5$

$9 \Rightarrow$ Touches
 $-2 \Rightarrow$ Crosses
 $-5 \Rightarrow$ Crosses

52) $f(x) = -(x-2.4)(x-4)^3(x-e)^8$

$2.4 \Rightarrow$ Crosses
 $4 \Rightarrow$ Crosses
 $e \Rightarrow$ Crosses